

→ Types of events :-

① Independent events :- In probability, Independent events are events where the occurrence of one event does not effect the occurrence of other.

→ Mathematically, two events A & B are independent :

$$P(A \cap B) = P(A) \cdot P(B)$$

→ This means the probability of both events happening together is simply the product of their individual probabilities.

Ex :- Tossing a coin :-

→ Let A be the event that the first toss is heads.

→ Let B be the event that the second toss is Tails.

→ These two events are independent because the result of the first toss does not affect the second.

$$\begin{aligned} \Rightarrow [P(A \cap B) &= P(A) \cdot P(B)] & P(A) &= \frac{1}{2}, P(B) = \frac{1}{2} \\ &= \frac{1}{2} \times \frac{1}{2} & &= \frac{1}{4} \end{aligned}$$

Dependent Events :- In probability Dependent events are events where the occurrence of one event is affected the probability of the other.

Mathematically, two events A & B are dependent if

$$[P(A \cap B) \neq P(A) \cdot P(B)]$$

$$[P(B|A) \neq P(B)]$$

Ex: Drawing cards without Replacement :-

Suppose you have a deck of 52 cards.

Let A be the event of drawing an Ace, on the first draw.

Let B be the event of drawing another Ace, on the second draw without putting first card back.

Since the card is not replaced, the total number of cards reduces to 51.

So Probability of B is affected by A. They are dependent.

$$P(A \cap B) = P(A) \times P(B|A)$$

$$= \frac{4}{52} \times \frac{3}{51}$$

→ Mutually exclusive events :-

→ Mutually exclusive events are events that cannot happen at the same time.

→ Two events A & B are mutually exclusive if -

$$P(A \cap B) = 0$$

→ That means the probability that Both A & B happen together is 0.

Ex:- Tossing a Coin :-

- Event A → getting heads
- Event B → getting Tails.

You can't get both at the same toss, So A & B are mutually exclusive.

→ Rolling A Die :-

- Event A → Getting a 3
- Event B → getting a 5

A Single die can show only one event - both can't happen together. So these events are mutually exclusive.